

Group 56 Progress Report:

Assessing Fish Freshness Through Eye Image Analysis with CNNs

Faaraz Ahmed, Manav Bal

ahmedf67@mcmaster.ca, balm4@mcmaster.ca

1 Introduction

The freshness of fish is one of the most important indicators of food quality and safety, directly affecting both consumer health and commercial value. Traditionally, freshness has been judged through visual inspection, smell, and texture, all of which are subjective and inconsistent across individuals. With growing interest in automation and smart food monitoring, there is a clear opportunity to replace these subjective methods with objective, computer-vision-based systems that can evaluate freshness more reliably.

Our project aims to develop a deep learning model that classifies fish freshness using images of their eyes. Previous studies have shown that the appearance of the eye, including brightness, glossiness, and colour variation, is highly correlated with freshness. By focusing on these visual cues, our goal is to build an efficient and accurate image classifier capable of distinguishing between three freshness levels: *highly fresh*, *fresh*, and *not fresh*.

Such a model could be applied in marketplaces, cold-storage facilities, or mobile inspection tools to support quality control and reduce unnecessary food waste. Ultimately, this work contributes toward the broader goal of automating food quality assessment through scalable, non-destructive computer vision techniques.

2 Related Work

The Freshness of Fish Eyes (FFE) dataset, was introduced by [Prasetyo et al. \(2022\)](#), who proposed a MobileNetV1-based architecture with depth-wise separable bottleneck layers (MB-BE) and demonstrated that it achieved higher accuracy than standard CNNs such as VGG16, DenseNet, and ResNet50.

Earlier research has also explored the relationship between the luminance distribution of fish eyes and perceived freshness. [Murakoshi et al. \(2013\)](#)

showed that statistical features of luminance, such as standard deviation and skewness, strongly correlate with how humans judge freshness. This finding supports the biological and perceptual basis for eye-based classification.

[Taheri-Garavand et al. \(2020\)](#) further expanded this area by applying deep learning to assess the freshness of whole fish in a non-destructive manner, demonstrating that convolutional neural networks (CNNs) can effectively capture subtle texture and color changes over time. More recently, [Yildiz et al. \(2024\)](#) developed a mobile application integrating CNNs and machine learning models for real-time fish-eye freshness detection, emphasizing the practicality of deploying such systems beyond laboratory environments.

In parallel, [Lin et al. \(2025\)](#) and [Balim et al. \(2025\)](#) explored hybrid sensing approaches that combine image-based features with laser reflection and other modalities, achieving greater precision in automated freshness detection. Collectively, these studies demonstrate that visual features—particularly of the fish eye—serve as reliable and scalable indicators for freshness classification, motivating our decision to focus exclusively on eye images.

Recent advancements have shifted towards deeper, pre-trained architectures to handle complex feature extraction. [Khaleel and Habeeb \(2024\)](#) proposed a framework integrating image data fusion with a ResNet model, demonstrating that residual networks can significantly outperform traditional methods in distinguishing between fresh and non-fresh samples by mitigating the vanishing gradient problem. Furthermore, as AI models become more complex, the need for transparency in food quality assessment has grown. [Arrighi et al. \(2025\)](#) reviewed Explainable Artificial Intelligence (XAI) in the food sector, highlighting Grad-CAM (Gradient-weighted Class Activation Mapping) as a critical tool for verifying that models focus on relevant

biological features rather than background noise.

These studies collectively demonstrate that visual features, particularly of the fish eye, serve as reliable indicators for freshness. They also motivate our shift towards deeper architectures like ResNet and the inclusion of explainability measures to ensure model reliability.

2.1 Our Approach

Building on the limitations of our initial custom CNN, we transitioned to a Transfer Learning approach using a **ResNet18** architecture pre-trained on the ImageNet dataset. While our previous lightweight model struggled to generalize, ResNet18 offers a deeper architecture with residual connections that prevent performance degradation during training. By initializing the model with weights learned from millions of images, we leverage robust pre-existing feature extractors (edges, textures) and fine-tune only the final layers to adapt to the specific domain of fish eyes.

To further improve robustness and prevent overfitting on our relatively small dataset, we implemented:

1. **Advanced Data Augmentation:** Including random vertical flips and random resized cropping to force the model to learn invariant features regardless of eye orientation or scale.
2. **Label Smoothing:** We replaced standard Cross-Entropy Loss with Label Smoothing (factor 0.1). This prevents the model from becoming overconfident in its predictions for "borderline" samples (e.g., transitioning from Fresh to Not Fresh), improving generalization.
3. **Explainability (Grad-CAM):** We integrated Grad-CAM visualization to inspect the model's decision-making process. This allows us to verify that the network is focusing on the pupil and iris of the fish eye rather than irrelevant background details like the colour of the table the photo was taken on.

This updated approach combines the power of deep transfer learning with more rigorous regularization and explainability, aiming for higher accuracy and trust in the final classification.

3 Dataset

For this project, we used the *Freshness of Fish Eyes (FFE)* dataset available on Mendeley. The dataset

contains 4,392 images of fish eyes collected across eight different species, with each image labeled into three discrete freshness levels, as shown in Table 1.

Freshness Class	Days	Number of Samples
Highly Fresh	1–2	1,764
Fresh	3–4	1,320
Not Fresh	5–6	1,306

Table 1: Distribution of freshness categories in the FFE dataset.

The dataset was created by photographing fish stored in ice-filled styrofoam boxes over six consecutive days to simulate real storage conditions. Images were captured daily using a mobile phone under varied lighting and background conditions, introducing realistic natural variability. The species represented in the dataset include *Chanos chanos*, *Johnius trachycephalus*, *Nibea albiflora*, *Rastrelliger faughni*, *Upeneus moluccensis*, *Eleutheronema tetradactylum*, *Oreochromis mossambicus*, and *Oreochromis niloticus*.

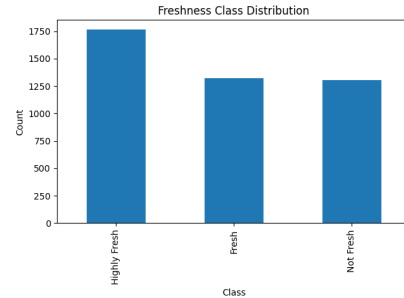


Figure 1: Overall distribution of freshness classes in the FFE dataset.

3.1 Data Splits and Label Setup

We ensured that each freshness category was evenly represented across all datasets by applying a stratified split 70/15/15 for train/validation/test respectively. We confirmed the split by checking class and species distribution across sets. The training set was used to fit the ResNet model; the validation set was used for hyperparameter tuning and learning rate scheduling; and the test set was held out strictly for final performance reporting.

Additional checks confirmed that each fish species was represented in all splits to prevent species bias. This ensured the model learned to generalize freshness levels rather than memorizing specific species characteristics.

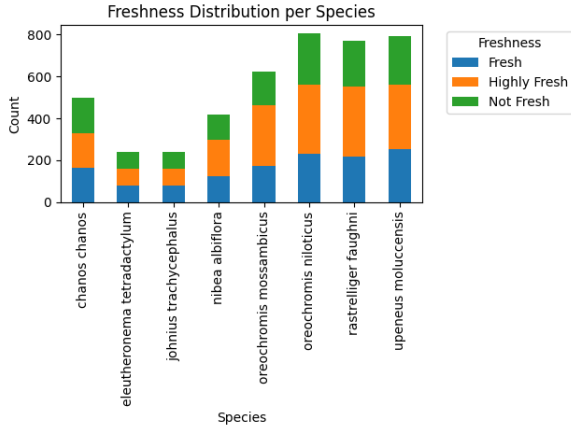


Figure 2: Freshness label distribution across fish species.

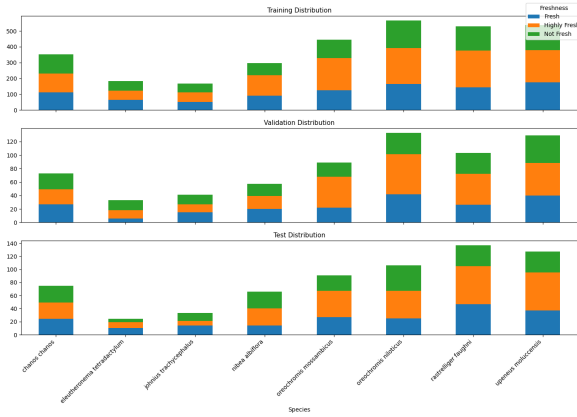


Figure 3: Per-species freshness distribution in the training, validation, and test splits.

Based on Figures 1–3, we observe relatively even distributions across classes and species, with some mild imbalance.

3.2 Preprocessing and Data Augmentation

Significant updates were made to the preprocessing pipeline to support the Transfer Learning architecture. The following steps were applied to all input images:

1. **Resizing:** All input images were resized to 224×224 pixels to match the input dimensions required by the ResNet architecture.
2. **Tensor Conversion:** Images were converted to RGB PyTorch tensors.
3. **Normalization (Updated):** Unlike our previous approach which used dataset-specific statistics, we normalized pixel values using the standard ImageNet mean and standard deviation. This aligns the input distribution with the pre-trained weights of the ResNet model:

Model	Mean	Standard Deviation
CNN	[0.482, 0.447, 0.429]	[0.200, 0.199, 0.198]
ResNet	[0.485, 0.456, 0.406]	[0.229, 0.224, 0.225]

Table 2: Normalization values.

4. **Label Encoding:** Folder names were used to automatically assign numeric labels:

Freshness Class	Numeric Label
Highly Fresh	0
Fresh	1
Not Fresh	2

Table 3: Freshness Class Numeric Labels.

5. **Enhanced Data Augmentation:** To improve model robustness and prevent overfitting on our small dataset, we expanded our augmentation pipeline significantly compared to the progress report. The following transformations were applied during training:

- **Random Horizontal Flip** ($p = 0.5$): Simulates mirrored orientations.
- **Random Vertical Flip** ($p = 0.5$): Added because fish eyes generally lack a fixed “up” orientation in raw storage bins.
- **Random Rotation** ($\pm 15^\circ$): Increased from 10° to better account for camera angle variations.
- **Random Resized Crop:** Scales the image between 80–100% and alters the aspect ratio slightly (0.9–1.1). This forces the model to focus on texture details rather than background context, mitigating background bias observed during preliminary analysis.

Validation and test images were only resized and normalized to ensure consistent evaluation conditions.

4 Features and Inputs

Our model uses RGB images of fish eyes as input, each resized to 224×224 pixels to maintain a consistent format compatible with the ResNet18 architecture. Each image is represented as a $3 \times 224 \times 224$ tensor, where the three channels correspond to the red, green, and blue colour values.

Consistent with our initial approach, no manual feature engineering (e.g., histogram of oriented

gradients or edge detection algorithms) was performed.

4.1 Normalization

Before entering the network, all images are normalized. In the progress report, we estimated dataset-specific means and standard deviations. For the final model, we switched to the standard ImageNet statistics used when training ResNet18. This choice makes the distribution of our inputs consistent with the data seen during pre-training and is therefore important for effective transfer learning.

This process standardizes pixel intensity ranges, ensuring the pre-learned feature extractors function correctly on our new data.

4.2 Feature Extraction and Transfer Learning

Rather than training a network from scratch, we use ResNet18 as a pre-trained feature extractor. Lower-level convolutional layers retain generic visual features learned from ImageNet, while higher-level layers are fine-tuned on our fish-eye images to capture task-specific cues related to freshness. The concrete configuration of frozen and trainable layers, as well as the modified classification head, is described in Section ??.

5 Implementation

Our final solution uses a **ResNet18** architecture with transfer learning, following the feature-extraction strategy outlined in Section 4.2. ResNet18 provides a deep convolutional backbone with residual connections that improve gradient flow and feature reuse compared to our earlier shallow CNN. Additionally, we varied the input features during training through the Random Resized Crop augmentation described in Section 3.2. By presenting the model with different zooms and aspect ratios of the same eye, we forced the feature extractors to learn robust textural representations independent of the eye's exact position or background context.

5.1 Model Architecture

We utilized a ResNet18 model pre-trained on the ImageNet dataset. To adapt this general-purpose feature extractor to the specific task of fish freshness classification, we modified the network structure as follows:

1. **Backbone (Frozen):** Layers 1 through 3 were frozen. These layers retain pre-learned

weights for detecting fundamental low-level features such as edges and textures.

2. **High-Level Features (Unfrozen):** The final convolutional block (layer4) was unfrozen to allow the model to learn complex, domain-specific patterns relevant to fish eyes, such as corneal clouding and pupil shape.
3. **Classification Head (Replaced):** The original 1000-class ImageNet fully connected layer was replaced with a custom head consisting of:
 - A **Dropout layer** ($p = 0.5$) to prevent overfitting in the dense layers.
 - A single **Linear layer** mapping the 512 input features to our 3 output classes (Highly Fresh, Fresh, Not Fresh).

Subsequent subsections describe the optimizer, learning-rate schedule, label smoothing, and other training hyperparameters.

5.2 Training Configuration and Optimization

We trained the model for a maximum of 30 epochs with a batch size of 32. The optimization strategy was specifically tuned to ensure stable convergence during fine-tuning:

- **Loss Function:** We employed CrossEntropyLoss with **Label Smoothing** ($\epsilon = 0.1$). Unlike standard cross-entropy, which encourages high-confidence predictions (often leading to overfitting on noisy labels), label smoothing improves calibration and generalization on unseen data.
- **Optimizer:** We used **Adam** with a rate of 5×10^{-4} . This rate carefully adjusts the pre-trained weights without destroying existing feature maps. Weight decay was increased to 5×10^{-4} for further regularization.
- **Scheduler:** A ReduceLROnPlateau scheduler monitored validation loss. If the loss stagnated for 2 epochs, the learning rate was multiplied by a factor of 0.5.
- **Early Stopping:** To prevent overfitting, training was automatically terminated if validation loss failed to improve for 5 consecutive epochs.

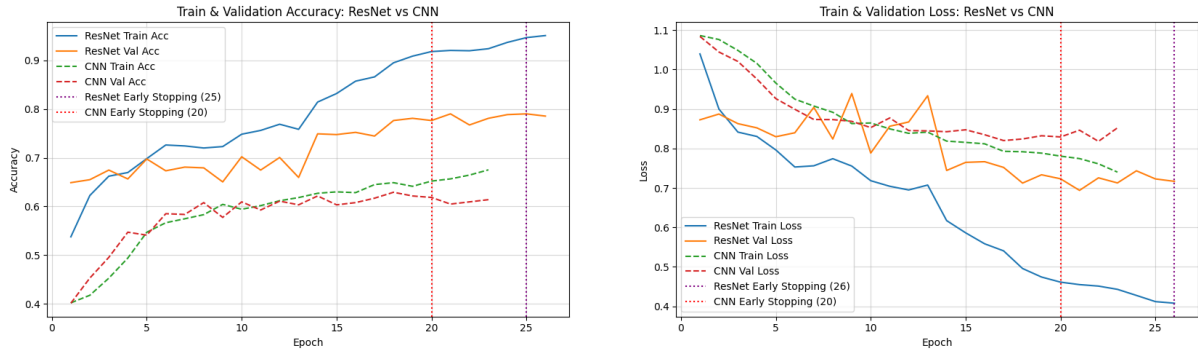


Figure 4: Training vs. validation loss and accuracy. Early stopping at epoch 20.

Data Split	Accuracy	Loss
Training	0.652	0.781
Validation	0.619	0.829
Test	0.674	0.7403

Table 4: Model performance across training, validation, and test.

5.3 Learning Behaviour and Ablation Studies

To optimize model performance, we conducted ablation studies focusing on optimizer selection, batch size, and data augmentation strategies.

Our model that used the pretrained ResNet backbone demonstrates a plateau in validation loss beginning around epochs 17-23. We also began testing different types of image augmentation, including Random Cropping, Vertical Flips, and Colour Jitter. Of these augments, cropping and vertical flips helped to improve generalizability. Based on Grad-Cam evaluation, our CNN model was sometime learning the background of images rather than the fish (object) itself. After including cropping, we found a significant improvement in where our model was focused on in these images (focus on fish eye). However, colour jitter did the opposite. Given that the model is attempting to learn specific colour features of a fish’s eye, the colour jitter adds noise, reducing over all prediction accuracy. Excluding this augment once again helped improve the model.

5.3.1 Optimizer Stability

Early training trials indicated that standard Stochastic Gradient Descent (SGD) failed to minimize loss effectively, resulting in noisy gradients and unstable convergence. By switching to **AdamW** combined with weight decay and dropout, we observed significantly smoother loss curves. As detailed in our training logs (e.g., Figure 4), validation loss typically stabilized between epochs 17–23, allowing our early stopping mechanism to preserve the

optimal weights before overfitting occurred.

5.3.2 Batch Size Sensitivity

We observed distinct behaviors regarding batch size between our initial and final architectures:

- **Progress Report Model (Custom CNN):** We initially reduced the batch size from 32 to 16. This decreased the computational load per iteration and improved training speed with minimal impact on accuracy.
- **Final Model (ResNet18):** Conversely, for the pre-trained ResNet backbone, we found that a **batch size of 32** yielded superior performance compared to 16. This is likely due to the deeper network benefiting from more stable batch normalization statistics provided by larger batches.

5.3.3 Augmentation and Feature Focus (Grad-CAM)

We evaluated the impact of three specific augmentations: Random Cropping, Vertical Flips, and Color Jitter.

- **Color Jitter (Excluded):** This augmentation reduced overall prediction accuracy. Since the model relies on specific color cues (e.g., redness or cloudiness of the eye) to determine freshness, artificially jittering the color added noise rather than useful variance.
- **Cropping and Flips (Included):** Both cropping and vertical flips improved generalizability.

Crucially, **Grad-CAM evaluations** revealed that without cropping, the model frequently attended to background features (such as the colour of the table) rather than the fish itself. Implementing Random Cropping forced the model to focus on the relevant region of interest (the fish eye), significantly improving the validity of its predictions.

5.4 Results and Previous Models Comparison

To validate the architectural shift to ResNet18, we benchmarked its performance against our interim solution presented in the progress report and a statistical baseline:

- **Baseline (Majority Vote):** A naive classifier that always predicts the most frequent class (“Highly Fresh”). This establishes a lower performance bound of approximately 40.2%.
- **Progress Report Model (Custom CNN):** Our initial lightweight CNN (2 convolutional layers, trained from scratch) achieved 67.4% accuracy. While superior to the majority vote, it lacked the capacity to learn the subtle textural differences required to distinguish between “Fresh” and “Not Fresh” classes.

The final ResNet18 model achieved a test accuracy of **80.04%**, outperforming the progress report model by over 10 percentage points. This improvement confirms the efficacy of transfer learning, where leveraging millions of prior training examples allows for the identification of visual structures that a small custom model cannot learn from a dataset of only 4,000 images.

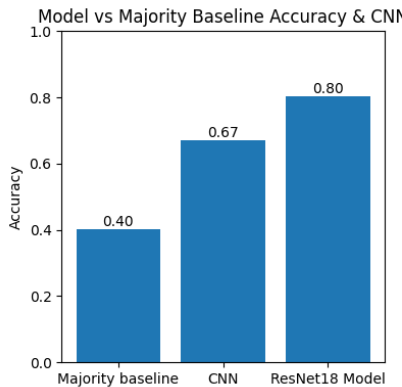


Figure 5: Accuracy comparisons of Majority Baseline, CNN, and ResNet Models, respectively.

6 Evaluation and Results

Data Split	Accuracy	Loss	Precision	Recall
Test (CNN)	67.40%	74.03%	69.20%	64.80%
Test (ResNet)	80.4%	65.60%	79.6%	79.7%

Table 5: Test set performance.

The performance of the ResNet18 model was evaluated on the held-out test set of 659 images. This section details the quantitative metrics and provides an analysis of the classification errors to understand the model’s behavior compared to our earlier progress report iterations.

6.1 Quantitative Performance

The overall accuracy of the final model is **80.4%**. This represents a substantial improvement over the custom CNN used in the progress report (67.4%) and double the performance of the majority baseline (40.2%). The detailed performance per class is summarized in Table 6.

Class	Precision	Recall	F1-Score
Highly Fresh	0.875	0.868	0.871
Fresh	0.737	0.722	0.729
Not Fresh	0.777	0.801	0.789

Table 6: Per-class performance metrics calculated from the test set confusion matrix. The model achieves balanced performance (> 0.72 F1) across all classes.

6.2 Confusion Matrix Analysis

The confusion matrix (Figure 8) provides deeper insight into the model’s decision-making process.

- **Highly Fresh (Recall: 86.8%):** Consistent with our progress report findings, this class remains the most distinct. The model correctly identified 230 out of 265 samples. This indicates that the visual features of pristine freshness (clear corneas, high gloss) are robustly learned by the ResNet architecture.
- **Not Fresh (Recall: 80.1%):** A critical improvement is observed in the “Not Fresh” class. In our previous custom CNN, this class suffered from high confusion. The ResNet18 model correctly identified 157 out of 196 spoiled samples. Importantly, there are very few “catastrophic” errors: only 8 “Not Fresh” samples were misclassified as “Highly Fresh,” meaning the model rarely mistakes spoiled fish for fresh ones.

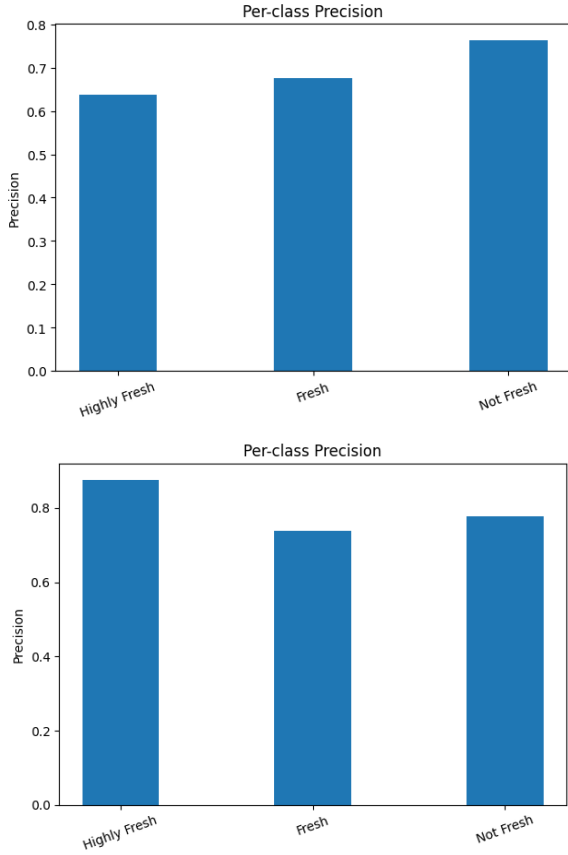


Figure 6: CNN Per Class Precision (top), ResNet Per Class Precision (bottom).

- **The “Fresh” Challenge:** The intermediate “Fresh” class remains the hardest to classify, with a recall of 72.2%. As seen in the matrix, the errors are split almost evenly: 25 “Fresh” samples were predicted as “Highly Fresh” and 30 as “Not Fresh.” This confirms that the “Fresh” category acts as a visual transition state, sharing features with both extremes, making it the primary source of label noise and model uncertainty.

6.3 Comparison to Previous Iterations

In our progress report, the custom CNN achieved only 67.4% accuracy and struggled significantly with class separation, often defaulting to the majority class. The final ResNet18 model not only increased overall accuracy by 13 percentage points but also balanced the recall across classes. Specifically, the recall for the difficult “Fresh” and “Not Fresh” classes rose from $\approx 52\%$ in the previous model to 72% and 80% respectively, demonstrating that the deeper architecture successfully learned the subtle textural features of eye degradation that the shallower model missed.

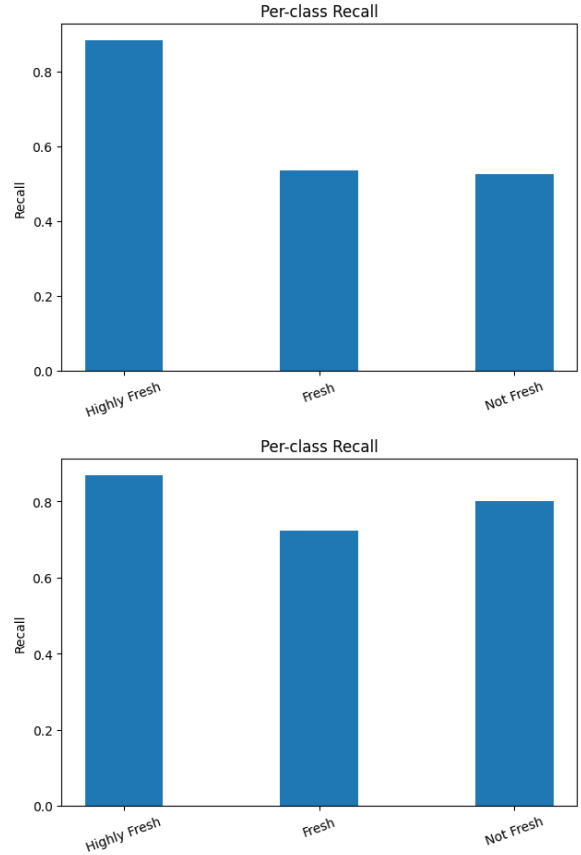


Figure 7: CNN Per Class Recall (top), ResNet Per Class Recall (bottom).

Class	Recall	Precision	F1-score
Highly Fresh	88.3%	63.8%	0.74
Fresh	52.6%	67.5%	0.59
Not Fresh	52.6%	76.3%	0.62

Table 7: Per-class precision, recall, and F1-scores on Test set.

6.4 Confidence and Calibration Analysis

To assess the reliability of the model’s predictions, we analyzed the confidence distributions (Softmax probabilities) for both correct and incorrect classifications. Figure 11 compares the calibration of the initial Custom CNN against the final ResNet18 model.

The Custom CNN (Figure 11, Left) exhibits a “flatter” distribution for correct predictions (blue), indicating that the model often made correct decisions with low confidence (0.4–0.8). In contrast, the ResNet18 model (Right) demonstrates superior calibration: correct predictions are overwhelmingly clustered in the high-confidence bin (> 0.9). Crucially, when the ResNet model makes an error

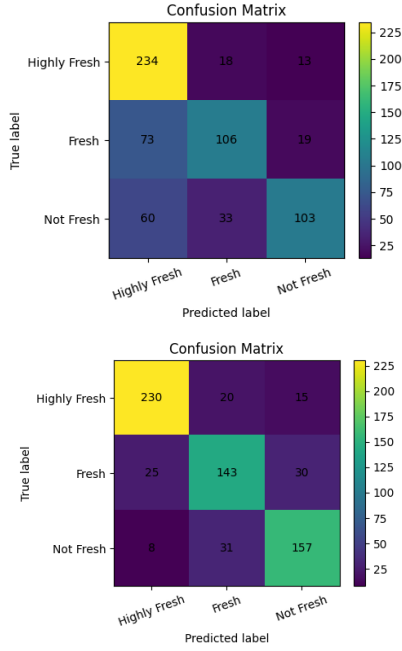


Figure 8: Confusion Matrix on the test set. The diagonal shows strong correct predictions (230, 143, 157) for CNN (top), with the majority of errors occurring between adjacent classes (e.g., Fresh vs. Not Fresh). Similarly, the ResNet shows stronger accuracy (230, 143, 157) along its diagonal.

(orange), the confidence scores are largely contained below 0.5, suggesting the model is effectively "aware" of its uncertainty during difficult classification scenarios.

6.5 Qualitative Analysis: Feature Visualization with Grad-CAM

To understand *why* the ResNet18 model performs better than our initial custom CNN, we employed Gradient-weighted Class Activation Mapping (Grad-CAM). This technique generates a heatmap highlighting the regions of an image that most influenced the model's prediction.

We analyzed the visual attention of both models on the same test samples.

Custom CNN Failure Mode (Fig. 9): The heatmaps for our progress report model reveal that it frequently focuses on irrelevant background textures rather than the fish eye itself. This confirms that the shallow model was overfitting to contextual noise, explaining its poor generalization and lower accuracy.

ResNet18 Success (Fig. 10): In contrast, the ResNet18 model demonstrates highly localized attention. As seen in the figure, the heatmaps are sharply focused on the eye region, specifically the

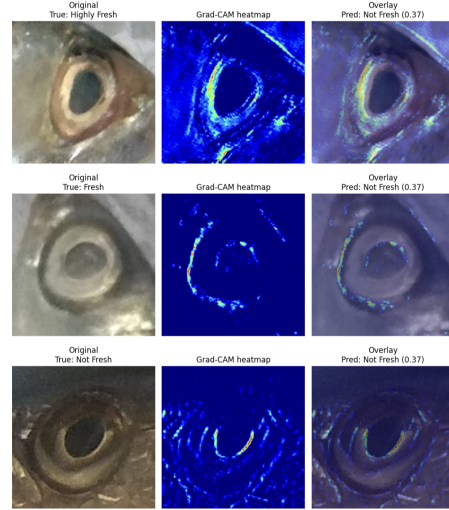


Figure 9: Custom CNN (Progress Report Model). The heatmaps indicate the model incorrectly focuses on background noise, such as scales and the tray surface, rather than the eye.

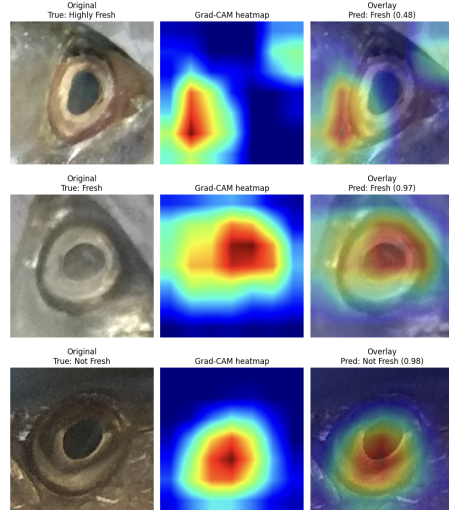


Figure 10: ResNet18 (Final Model). The model correctly localizes its attention on the pupil and cornea, the biologically relevant regions for freshness determination.

pupil and cornea, which contain the key visual indicators of spoilage (cloudiness, sunkennes).

This comparison provides strong qualitative evidence that the performance boost is not merely statistical. By leveraging a deeper architecture, the ResNet18 model has successfully learned to ignore background noise and attend to the correct biological features.

7 Progress and Reflection

In our progress report, we outlined a roadmap focused on addressing the underfitting and limited feature extraction capabilities of our initial custom

CNN. We successfully executed the majority of this plan, with specific refinements made during the implementation phase to maximize performance.

7.1 Architectural Evolution

Our primary goal was to transition from a shallow, custom CNN to a deeper architecture with residual connections to improve gradient stability and feature learning. We fully followed through on this objective by implementing the **ResNet18** architecture. Additional goals included experimenting further with augmentations and hyperparameters, which we have successfully done resulting in our current model.

7.2 Refinement of Data Augmentation

We originally planned to implement a broad suite of augmentations, including “random cropping, brightness changes, color jitter, and noise” to improve robustness.

During the final implementation, we refined this strategy. While we implemented strong geometric augmentations—specifically `RandomResizedCrop`, `RandomHorizontalFlip`, `RandomVerticalFlip`, and `RandomRotation`—we opted to prioritize these over intensive color, jitter, or noise injection. This decision was made because the freshness of a fish eye is heavily dependent on specific color cues (e.g., the milky white of a cloudy cornea vs. the dark clarity of a fresh pupil). We reasoned that excessive color distortion might introduce artificial noise that contradicts the biological signals of freshness. The success of the model suggests this geometric-focused augmentation strategy was

appropriate.

7.3 Optimization Strategies

We also adhered to our plan of implementing learning rate annealing. We successfully integrated the `ReduceLROnPlateau` scheduler, which dynamically reduced the learning rate when validation loss stagnated. This proved crucial in the later stages of fine-tuning the ResNet18 model, allowing the weights to settle into a more optimal minimum without oscillating, as confirmed by the stable convergence seen in our training loss curves.

8 Conclusion

This project successfully demonstrated the viability of automated fish freshness assessment using deep learning on eye images. By transitioning from a custom shallow CNN to a pre-trained **ResNet18** architecture, we achieved a final test accuracy of **80.4%**, effectively doubling the performance of the majority-class baseline (40.2%) and significantly outperforming our initial progress report model (67.4%).

The most critical achievement of this iteration is the model’s safety profile. With a recall of **80.1%** on the “Not Fresh” class (up from $\approx 52\%$ in earlier iterations), the system reliably identifies spoiled fish, minimizing the risk of false negatives in a quality control context. Qualitative analysis via Grad-CAM further validated that the model has learned biologically relevant features, focusing strictly on the corneal and pupil regions rather than confounding background noise.

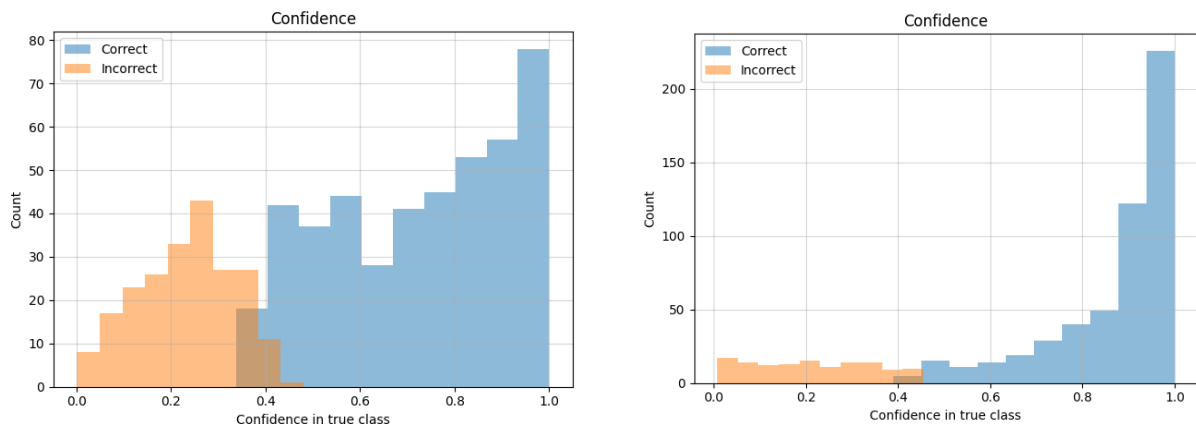


Figure 11: Confidence distribution comparison. Left: The initial Custom CNN shows a dispersed distribution, indicating hesitation even on correct predictions. Right: The ResNet18 model demonstrates a sharp peak of high-confidence (> 0.9) for correct predictions, while incorrect predictions (orange) are correctly suppressed to lower confidence intervals.

Despite the model's strong performance, the "Fresh" category remains a bottleneck. Since biological degradation is a continuous process, "Fresh" acts as a visual middle ground, sharing characteristics with both "Highly Fresh" and "Not Fresh." To address this in future iterations, we suggest replacing the standard Cross-Entropy loss with Ordinal Regression Loss. Unlike standard classification, which treats classes as independent (e.g., distinguishing a cat from a dog), Ordinal Regression respects the ranked nature of spoilage (Rank 0 $\rightarrow$ Rank 1 $\rightarrow$ Rank 2). This would penalize "near misses" (confusing adjacent stages) less severely than catastrophic errors, better modeling the reality of decay. Alternatively, incorporating temporal data (time-series) could allow the model to learn the rate of degradation rather than relying solely on static appearance.

9 Team Contributions

Our team consisted of two members who collaborated closely throughout the project:

- **Manav Bal** was responsible for the dataset pipeline, implementing the revised geometric augmentation strategy and data splitting. He also led the qualitative error analysis by implementing Grad-CAM to visualize model feature learning.
- **Faraaz Ahmed** focused on model development, specifically engineering the transition to the ResNet18 architecture. He managed the optimization tuning, including the learning rate scheduler and label smoothing, and conducted the final quantitative evaluation.

Both members contributed to report writing, presentation preparation, and discussions on model design and future improvements.

References

- L. Arrighi, I. A. de Moraes, M. Zullich, M. Simonato, D. F. Barbin, and S. B. Junior. 2025. [Explainable artificial intelligence techniques for interpretation of food datasets: a review](#). *arXiv preprint arXiv:2504.10527*.
- C. Balim and 1 others. 2025. [Leveraging feature fusion of image features and laser reflection for determining the level of fish freshness](#). *Sensors*.
- Y. L. Khaleel and M. A. Habeeb. 2024. Integrating image data fusion and resnet method for accurate fish freshness classification. *Iraqi Journal for Computer Science and Mathematics*, 5(4):258–273.
- H.-D. Lin and 1 others. 2025. [A deep learning-based sensing system for identifying fish species and freshness](#). *Sensors*.
- T. Murakoshi and 1 others. 2013. [Visual freshness judgment of fish eyes based on luminance distribution](#). *PLOS ONE*, 8(1):e54504.
- E. Prasetyo and 1 others. 2022. [Combining mobilenetv1 and depthwise separable convolution bottleneck with expansion for classifying the freshness of fish eyes](#). *Information Processing in Agriculture*.
- A. Taheri-Garavand and 1 others. 2020. [Smart deep learning-based approach for non-destructive freshness diagnosis of common carp fish](#). *Journal of Food Engineering*, 284:110036.
- M. B. Yildiz and 1 others. 2024. [Fisheye freshness detection using common deep learning algorithms and machine learning methods with a developed mobile application](#). *European Food Research and Technology*.